



Post Office Box 2000, Decatur, Alabama 35609-2000

May 29, 2025

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Unit 3
Renewed Facility Operating License No. DPR-68
NRC Docket No. 50-296

Subject: **Licensee Event Report 50-296/2025-001-01**

Reference: Letter from TVA to NRC, "Licensee Event Report 50-296/2025-001-00,"
dated April 9, 2025 (ML25099A030)

The enclosed Licensee Event Report provides details of an Automatic Reactor Scram Following Main Generator Trip. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(iv)(A), as an automatic actuation of the Reactor Protection System, the Primary Containment Isolation System, and the Emergency Diesel Generators. Revision 1 to this report includes information from the cause evaluation that was not previously reported.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact David J. Renn, Site Compliance Manager, at (256) 729-2636.

Respectfully,

A handwritten signature in black ink, appearing to read "D. A. Komm", is written over a horizontal line.

Daniel A. Komm
Site Vice President

Enclosure: Licensee Event Report 50-296/2025-001-01 – Automatic Reactor Scram
Following Main Generator Trip

cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant
NRC Project Manager - Browns Ferry Nuclear Plant

ENCLOSURE

**Browns Ferry Nuclear Plant
Unit 3**

Licensee Event Report 50-296/2025-001-01

Automatic Reactor Scram Following Main Generator Trip

See Enclosed

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 04/30/2027				
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)				Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: pira_submission@omb.eop.gov . The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.				
1. Facility Name Browns Ferry Nuclear Plant, Unit 3				☒ 050 ☐ 052		52. Docket Number 00296				
3. Page 1 OF 6										
4. Title Automatic Reactor Scram Following Main Generator Trip										
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
02	08	2025	2025	- 001 -	01	05	29	2025	N/A	05000XXX
									Facility Name	Docket Number
									N/A	05000XXX
9. Operating Mode 1						10. Power Level 070				
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)										
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)				☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)				
☐ OTHER (Specify here, in abstract, or NRC 366A).										
12. Licensee Contact for this LER										
Licensee Contact Denzel Housley, Compliance Engineer								Phone Number (Include area code) 256-729-7643		
13. Complete One Line for each Component Failure Described in this Report										
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
4. Supplemental Report Expected)					15. Expected Submission Date			Month	Day	Year
☒ No								N/A	N/A	N/A
☐ Yes (If yes, complete 15. Expected Submission Date										
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)										
On February 8, 2025, at 2248 Central Standard Time, while at approximately 70 percent rated thermal power, Unit 3 experienced a main generator trip causing an automatic reactor scram. Plant equipment responded as expected and Unit 3 was transitioned to Mode 4. The Unit 3 isophase bus electrical fault was most likely initiated by a combination of particulate buildup on standoff insulators and elevated humidity. The root cause of the event was that the station failed to maintain an adequate maintenance strategy for the bus duct system due to perceived system reliability. Repairs were made to the main generator bus which allowed the reactor to restart and sync to the electrical grid on March 6, 2025. Actions planned include revising the applicable Preventive Maintenance items and procedure to inspect and clean all insulators and ensure proper grounding connections.										

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 3	☒ 050	2. DOCKET NUMBER 00296	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 001	REV NO. - 01

NARRATIVE**I. Plant Operating Conditions before the Event**

At the time of this event, Browns Ferry Nuclear Plant (BFN) Unit 3 was in Mode 1 at approximately 70 percent Rated Thermal Power (RTP).

II. Description of Event**A. Event Summary**

On February 8, 2025, at 2248 Central Standard Time (CST), while at approximately 70 percent RTP, BFN Unit 3 experienced a main generator [EL] trip causing an automatic reactor scram [JC]. Plant equipment responded as expected and Unit 3 was transitioned to Mode 4.

During the scram response, there was a valid actuation of the Primary Containment Isolation System (PCIS) [JM] Groups 2, 3, 6, and 8. Upon receipt of these signals, components actuated as required. A loss of power to the 3EA and 3EB 4KV shutdown boards resulted in a valid actuation of the 3A and 3B emergency diesel generators (EDGs) [EK] which successfully re-energized the boards as designed.

The Tennessee Valley Authority (TVA) is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(iv)(A), as an automatic actuation of the Reactor Protection System (RPS), the PCIS, and the EDGs.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, systems, or components whose inoperability contributed to this event.

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1. FACILITY NAME

Browns Ferry Nuclear Plant, Unit 3



050



052

2. DOCKET NUMBER

00296

3. LER NUMBER

YEAR	SEQUENTIAL NUMBER	REV NO.
2025	- 001	- 01

NARRATIVE**C. Dates and approximate times of occurrences**

<u>DATE AND APPROXIMATE TIMES</u> (Central Time)	<u>OCCURRENCE</u>
February 8, 2025, at 2248	Unit 3 experienced a reactor scram following a main generator trip. (Event Notification report 57537 was reported to the NRC for this event.)
March 6, 2025	Following repairs to the unit 3 main generator bus, the unit was synced to the grid

D. Manufacturer and model number of each component that failed during the event

N/A

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error

The main generator trip was evidenced by the reactor scram.

G. The failure mode, mechanism, and effect of each failed component

The reactor trip was caused by a fault in the isophase bus [IPBU] which resulted in a main generator trip.

H. Operator actions

Operations personnel stabilized the plant following the reactor scram and subsequently initiated a plant cooldown to Mode 4.

I. Automatically and manually initiated safety system responses

PCIS Groups 2, 3, 6, and 8 isolation signals were received. Upon receipt of these signals, components actuated as required. Additionally, the 3A and 3B EDGs automatically started to re-energize their respective shutdown boards.

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	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 001	REV NO. - 01

NARRATIVE**III. Cause of the event****A. Cause of each component or system failure or personnel error**

The Unit 3 isophase bus electrical fault was most likely initiated by a combination of particulate buildup on standoff insulators and elevated humidity. The root cause of the event was that the station failed to maintain an adequate maintenance strategy for the bus duct system due to perceived system reliability.

Contributing causes to this event were 1) roles and responsibilities for Engineering and Maintenance were not clearly defined to ensure Doble Electromagnetic Interference (EMI) test reports were reviewed and acted upon, and 2) original isophase bus duct design does not allow access to all standoff insulators for inspection, cleaning, and replacement.

B. Cause(s) and circumstances for each human performance related root cause

N/A

IV. Analysis of the event

On February 8, 2025, the Browns Ferry Nuclear Plant Unit 3 main generator tripped on an electrical fault within the isophase bus duct which led to a reactor scram.

V. Assessment of Safety Consequences

The plant responded as designed, while maintaining defense-in-depth for nuclear safety. All nuclear safety systems functioned as designed. This event was of very low nuclear safety significance. At no time was the health and safety of the public at risk.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

N/A

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shut down.

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NARRATIVE**C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service**

There was no safety system rendered inoperable.

VI. Corrective Actions

Corrective Actions are being managed by the TVA corrective action program under condition report 1990732.

A. Immediate Corrective Actions

Following repairs to the main generator bus, the reactor was restarted and synced to the electrical grid on March 6, 2025.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future**Root Cause:**

- Scope will be added to the Preventive Maintenance items (PMs) for each unit to inspect and clean insulators from the generator to the transformers (1990732-032 and -033).
- The procedure (EPI-0-242-BUS001) used in each PM will be revised to reflect working all insulators (1990732-034); provide location diagrams (1990732-038, -039, and -040) of the insulators, links, and access hatches; add inspections of the airboxes for evidence of partial discharge or arcing; and replace cleaning solvent LPS (brand name), which leaves a residue, with isopropyl alcohol.
- Actions to ensure proper grounding connection and proper electrical isolation between bus duct sections will be added to EPI-0-242-BUS001. (1990732-020, -021, -032, -033, and -034).

Contributing Cause 1:

- The two-year PM that directs Doble EMI data gathering will be revised to ensure a CR is initiated and a review performed for the final report (1990732-028).

Contributing Cause 2:

- During the Unit 2 2025 refueling outage, access hatches were installed on the Unit 2 isophase bus duct between the generator and the Generator Circuit Breaker (GCB) allowing access to all insulators (1990732-014).
- A follow-up action will ensure access hatches are installed on the Unit 2 area between the GCB and the transformers (1990732-015).



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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NARRATIVE

- Access hatches for all the Unit 3 and Unit 1 insulators will be provided in upcoming refuel outages (1990732-010 and -013).

VII. Previous Similar Events at the Same Site

A search of LERs from BFN, Units 1, 2, and 3 over the last five years identified no similar events.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].